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Dataset Description

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Global Climate Change Dataset

Dataset Description & Feature Reference
Course: Business Intelligence & Data Warehousing

1. Dataset Overview

The Beginner Climate Change Dataset is a structured tabular dataset designed for introductory data analysis, machine learning, and business intelligence projects. It captures key environmental, energy, and climate-risk indicators across 15 countries from 1990 to 2024.

Attribute	Value
Total Rows	1,200
Total Features / Columns	20 (14 original + 6 identifier / derived)
All Countries Covered	15 — Pakistan, India, China, USA, Germany, Brazil, Australia, Nigeria, Russia, Japan, France, Canada, South Africa, Mexico, Indonesia
Time Period	1990 – 2024
Granularity	One record per country per year
Data Format	CSV / PostgreSQL relational table
Source	Kaggle

2. Feature Reference Table

The table below describes all features in the dataset, including their data types, value ranges, and meanings.

Feature Name	Data Type	Range / Values	Description
year	Integer	1990 – 2024	Calendar year of the record
country	String	15 countries	Name of the country (e.g., Pakistan, USA, Brazil)
global_avg_temperature	Float	~12 – 17 °C	Global average surface temperature in degrees Celsius for that country-year
co2_concentration_ppm	Float	~320 – 440 ppm	Atmospheric CO_2 concentration in parts per million
heatwave_days	Integer	0 – 90	Number of days per year classified as a heatwave event
flood_events_count	Integer	0 – 50	Total number of significant flood events recorded in the year
drought_index	Float	0.0 – 1.0	Standardised drought severity (0 = no drought, 1 = extreme)
sea_level_rise_mm	Float	~1 – 6 mm/yr	Annual sea-level rise measured in millimetres
renewable_energy_share	Float	5 – 85 %	Percentage of total energy from renewable sources
fossil_fuel_consumption	Float	~200 – 800 PJ	Total fossil-fuel energy consumption in petajoules
air_quality_index	Float	20 – 200	Air Quality Index; values above 100 are considered unhealthy
forest_cover_percent	Float	5 – 80 %	Percentage of land area covered by forests
deforestation_rate	Float	0 – 5 % /yr	Annual rate of forest loss as a percentage of forest cover
climate_risk_index	Float	0.1 – 1.0	Composite climate risk score; values above 0.7 = high risk

3. Derived / Engineered Features

Two additional columns are computed by the ETL pipeline during the Transform stage and loaded into the data warehouse fact table:

Derived Feature	Formula & Purpose
temp_anomaly	global_avg_temperature − global mean temperature. Measures how much a country's temperature deviates from the worldwide average.
extreme_event_score	Weighted composite of heatwave_days + flood_events_count + drought_index. Summarises overall extreme-weather burden in a single KPI score.

4. Missing Values & Data Quality

The dataset contains a small number of intentional null values (approximately 15 per column) in the following three columns to simulate real-world data gaps:

- air_quality_index
- renewable_energy_share
- drought_index

During the ETL cleaning stage, all null values are imputed using the column median. Outliers are capped using the 1.5 × IQR (Inter-Quartile Range) method to prevent extreme values from distorting downstream analysis.

5. Data Warehouse Star Schema

After ETL processing, the dataset is restructured into a star schema within a PostgreSQL warehouse. The schema consists of one central fact table and three dimension tables:

Table	Type	Key Columns
fact_climate	Fact Table	time_id, country_id, energy_id + all 14 measures + 2 derived KPIs
dim_time	Dimension	time_id, year, decade, era, is_current_decade
dim_country	Dimension	country_id, country, region, is_developing
dim_energy	Dimension	energy_id, energy_type, description, is_clean

6. Potential Use Cases

- Regression analysis to predict global average temperature or climate risk index
- Classification to identify high-risk vs. low-risk countries (climate_risk_index > 0.7)
- Time-series analysis of CO_2 concentration and sea-level rise trends over three decades
- Clustering countries by environmental profile and energy mix
- Power BI / Tableau dashboard development for climate KPI reporting
- Star-schema data warehouse design and OLAP query practice

7. Technologies Used

Layer	Technology
Source Database	PostgreSQL (public.fact_climate_global)
ETL Language	Python 3.x
ETL Libraries	pandas, numpy, psycopg2
Data Warehouse	PostgreSQL — warehouse schema (star schema)
BI / Reporting Tool	Microsoft Power BI Desktop
DAX Measures	Avg Temperature, Total CO_2, Climate Risk Index, High Risk Count
Geospatial Visual	Azure Maps (Power BI built-in connector)